Technical report - Sales analysis

Company: VisionMetrics Analytics

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Year of project:2025.

1. Objective of the Analysis

VisioMetrics Analytics is a Costa Rican company dedicated to importing and selling technology products in the local market.

The main objective of this analysis is to:

Evaluate the 2024 sales performance and profitability by product, product category, city, and sales representative, with special focus on products with high Cost of Goods Sold (COGS) and potential net losses.

This work supports the company in understanding which products, markets, and sales profiles contribute most to profit, and which ones are structurally less profitable.

2. Methodology

The analysis was conducted using the 2024 sales database provided by the company in CSV format. The data was formatted, cleaned, and explored using Power Query in Excel, including the standardization of product, category, city, and salesperson fields. Based on this prepared dataset, several pivot tables were created to segment and analyze sales, COGS, gross profit, and net loss by product, product category, city, and sales representative. From these structures, three interactive dashboards were developed in Excel, using a cool color palette (blue, black, violet, light blue): an executive overview of sales and profitability, a commercial performance dashboard by category, city and sales representative, and a losses and risk monitoring dashboard focused on non-profitable products and high COGS patterns. After testing the dashboards to ensure correct calculations and interactions, key patterns and trends were identified and consolidated into this concise technical report, summarizing the main findings in a format suitable for management and stakeholders.

3. Key Findings

a) Global Cost Structure and Profitability:

COGS account for approximately 73% of total sales, leaving a gross margin of about 27%. Despite this margin, the company currently reports a net loss of around 8% of total sales, indicating that the existing cost structure and loss-generating items are creating an operational deficit.

b) Performance by Product Category:

The categories “Sensores,” “Seguridad,” and “Energía” are the top performers, consistently contributing the most both in sales volume and profit. In contrast, the “Asistentes” and “Redes” categories concentrate a significant portion of the losses, positioning them as structurally weaker lines from a profitability standpoint

c) Geographic Distribution and Market Concentration:

San José is the primary customer hub, concentrating the highest number of customers, sales, and gross profit. At the same time, San José also records the highest net loss in absolute value, suggesting that highly profitable products and deficit-generating products coexist in this market. Other cities show a low average number of customers (approximately 3 per city), indicating low market penetration or high geographic fragmentation outside the capital. There are differences in profitability between cities with similar or even lower sales volumes; for example, Cartago reports higher sales than Heredia, but Heredia generates approximately $2,000 more in profit than Cartago. Alajuela and Puntarenas display very similar levels of sales, costs, and profit. While San José has the largest deficit, the remaining cities show average net losses of around $5,000.

d) Sales Team Performance:

Fernanda Montenegro García is the top performer in both sales volume and profit contribution, representing a strong commercial profile. Gael Castro Silva has the lowest sales volume, with an approximate gap of $10,000 in sales compared to the top performer; the rest of the team is distributed between these two extremes. A notable case is Jimena Valverde Suárez, who is second in sales volume but generates the lowest total profit among the team. This suggests that the product mix in her portfolio is less profitable, despite her strong sales volume.

e) Most Profitable Products (Top Performers)

Among the Top 5 products by profit, all of which do not generate net losses, the “Sensor de puerta” stands out as the highest profit contributor, even though it is only the second-highest product in sales volume. This product also has the lowest COGS within this top group, making it particularly attractive in terms of margin and profitability. Conversely, the “Cámara de seguridad WiFi” is the highest-selling product by volume, but ranks only fourth in total profit, and exhibits the highest COGS among the top profitable products, implying a tighter margin.

f) Products with Highest Losses and Risk

The “Router AX3000” is the single largest contributor to net losses: it is the second-highest selling item within the loss-generating group and has the highest COGS, making it particularly critical. The “Asistente de voz” and the “Interruptor Smart” are the next most relevant deficit-generating products. Two of these three high-loss products belong to the “Redes” and “Asistentes” categories, which confirms the weakness of these lines. The “Interruptor Smart” is part of the “Energía” category, showing that even within an overall strong category, certain individual products can be structurally unprofitable.

g) Losses by Product and Salesperson

When crossing net loss by product and salesperson, the analysis confirms that only three products generate the relevant net losses: Router AX3000, Asistente de voz, and Interruptor Smart. The average loss per unit for each of these products is highly consistent across all sales representatives, indicating that the sales team’s individual performance is not the main driver of these losses. The root cause of the deficit is primarily linked to the COGS and price structure of these products, rather than to how they are sold.

4.Deliverables

The main deliverables of this project are three interactive dashboards in Excel, a concise technical report, and optional annexes for visualization and sharing.

First, three interactive Excel dashboards were developed using a cool color palette (blue, black, violet, light blue). The Executive Sales & Profitability Dashboard – Business Overview and Key Financial Indicators (Sales 2024) provides a high-level view of total sales, COGS, gross profit, net loss and overall margin performance. The Commercial Performance Analysis – Sales, Profitability, and Market Distribution by Category, City, and Sales Representative focuses on how different product categories, geographic markets and sales representatives contribute to sales and profitability. Finally, the Losses and Risk Monitoring Dashboard – Identifying Non-Profitable Products, High COGS Patterns, and Improvement Opportunities is dedicated to highlighting non-profitable products, analyzing COGS behavior and showing how net losses are distributed across products and the sales team.

In addition to the dashboards, this technical report summarizes the objective, methodology and key findings in a compact format suitable for management review or client evaluation.

Fig 1: KPI Overview Dashboard

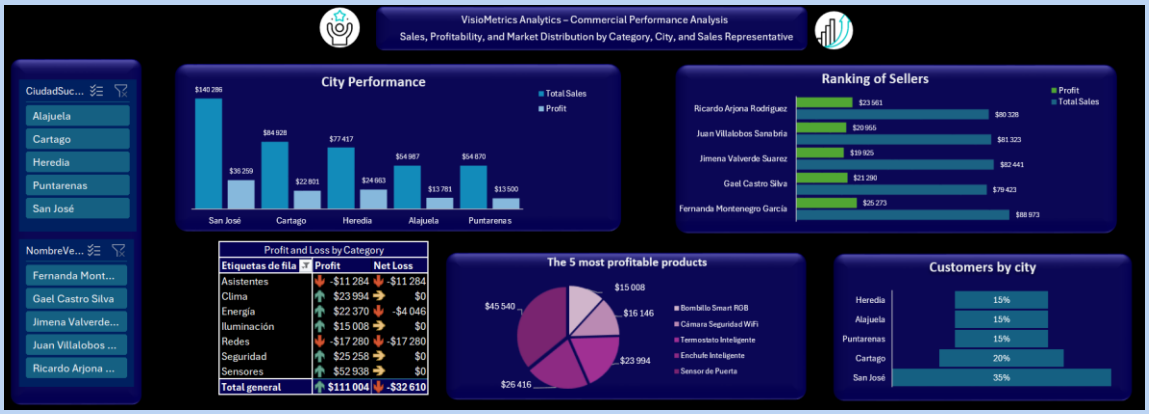

Fig 2: Sales Performance Review Dashboard

Fig 3. Loss and Risk Exposure Dashboard

Links to the dashboards: [Dashboard - VisioMetrics Analytics.xlsx]

The OneDrive link above is so that you can view and interact with the dashboard. If you have trouble interacting with the dashboard in your browser and want to see it work, you can download it and use it on your computer.

For security clarification purposes, the Excel file contains macro programming (VBA).